

Comprehensive Project Report for UNICOM TIC Management System

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1. Introduction

The UNICOMTIC Management System (UMS) is integrated software designed to streamline administrative processes across educational institutions. The system incorporates functionalities for managing students, staff, lecturers, and administrators, along with student-specific features such as marks, attendance, and timetables. This document outlines the project's design, development, and implementation details.

2. System Overview

The UMS offers a robust solution to handle multiple administrative tasks efficiently. Key features include:

- **Role-based User Management:** Secure login for administrators, staff, lecturers, and students.
 - **Student Management:** Adding, updating, retrieving, and deleting student data.
 - **Academic Management:** Handling marks, attendance, and timetables.
 - **Lecturer Assignment:** Assigning courses and subjects to lecturers.
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3. Problem Definition and Customer Requirements

Problem Definition

Institutions often face challenges in efficiently managing diverse academic and administrative activities. Manual processes are prone to errors, leading to inefficiencies and inaccuracies.

Customer Requirements

Functional Requirements:

- **Login and Registration:** Secure role-based authentication.
- **CRUD Operations:** Add, update, delete, and view records for students, staff, lecturers, and admins.
- **Student-Specific Management:** Features for marks, attendance, and timetable tracking.

Non-Functional Requirements:

- **Speed:** Real-time data processing.
 - **Responsive Design:** Adaptable to various devices.
 - **Security:** Data encryption and secure access.
 - **User-Friendly Interface:** Intuitive navigation and functionality.
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4. Feasibility Study

Technical Feasibility:

- **Software Requirements:** Windows 10+, .NET Framework 4.7+, Visual Studio 2022, SQLite.
- **Hardware Requirements:** Minimum Intel i3, 4 GB RAM, 500 GB storage.

Economical Feasibility:

- Budget allocation for software licenses and hardware procurement.

Legal Feasibility:

- Compliance with local regulations (e.g., .lk domain requirements).
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5. System Design

Use Case Diagram:

Illustrates interactions between system users (Admin, Lecturer, Student) and core features.

Entity-Relationship Diagram (ERD):

Defines relationships between entities like Students, Courses, Lecturers, and Subjects.

6. Features and Functionalities

Login with only login button for access based login

- Users able to login with username and password only, get access them role-based, automatically.

Student Management:

- Adding, updating, retrieving, and deleting student details.

Marks Management:

- Recording and retrieving marks per subject and student.

Attendance Tracking:

- Monitoring and managing student attendance records.

Timetable Management:

- Scheduling and viewing student timetables.

Staff and Admin Management:

- Handling staff and administrator data.

Lecturer Assignment:

- Assigning lecturers to courses and managing their schedules.
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7. Development and Coding Approach

Object-Oriented Programming (OOP):

- **Inheritance:** Common properties shared across entities.
- **Polymorphism:** Functionality tailored for specific user roles.

MVC Framework:

- Clean separation of concerns for efficient development and maintenance.
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8. Testing and Validation

Testing Overview:

- **Login Module:** All user roles successfully authenticated.
- **Database Operations:** CRUD operations tested and validated.
- **User Interface:** Navigation tested across all modules.
- **Error Handling:** Exceptions logged and handled gracefully.

Sample Test Case:

Test Case ID	Description	Expected Outcome	Actual Outcome	Status
TC001	Login as Admin	Redirect to Dashboard	Redirect to Dashboard	Pass
TC002	Add New Student	Student Added to DB	Student Added to DB	Pass
TC003	Update Student	Student Updated in DB	Student Updated in DB	Pass
TC004	View Student by access level	Student only access view	Student can view only in view mode	Pass
TC005	Input fields validation	Checks and notice input fields on typing	Checked and noticed about input fields.	Pass

Table 1Guantt Chart

The screenshot shows a web application interface with a sidebar menu on the left containing options like PERSONAL DETAILS, LECTURER, ADMIN, STAFF, COURSE SUBJECT, ATTENDANCE, TIME TABLE, and EXAMS / MARKS. The main content area displays a form titled 'OWN DETAILS' for user profile management. The form fields are as follows:

Field	Value
FULL NAME	vigetharan
UT / EMPLOYEE No	AD01001
ADDRESS	kopay
NIC NUMBER	863531330V
EMAIL	vige@ru.dl
SALARY	55000
CONTACT No.	0779112993
DATE OF BIRTH	12/18/1986 12:00:00 AM
GENDER	MALE
AGE	38
USER ROLE	ADMIN
DATE JOINED	2025-06-17

At the bottom of the form, there are two buttons: 'CANCEL' and 'UPDATE'.

9. Work Breakdown Structure

No	Task	Start.Date	End.Date	No of Days
01	Requirement gathering	10.06.2025	12.06.2025	02
02	System Design	13.06.2025	17.06.2025	04
03	Development	17.05.2025	20.06.2025	03
04	Testing	21.06.2025	22.06.2025	1
05	Deployment	22.06.2025	23.06.2025	1

Table 2Work BreakDown Structure

11. Recommendations

1. Implement a mobile application interface for enhanced accessibility.
 2. Add automated backup functionality for data.
 3. Add Automatic Attendance marking easily.
 4. Enhance reporting capabilities with graphical analytics.
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12. Conclusion

The UNICOM TIC Management System offers a comprehensive solution to streamline institutional processes. With its robust features and user-friendly design, UMS is well-equipped to meet modern academic and administrative demands.